Exercises learning task 11

2026 – 2027

Exercise 7.21: Let $X_1, X_2, \ldots, X_7$ denote a random sample from a population having mean μ and variance σ^2. Consider the following estimators of μ:

$$\hat{\Theta}_1 = \frac{X_1 + X_2 + \cdots + X_7}{7}$$

$$\hat{\Theta}_2 = \frac{2X_1 - X_6 + X_4}{2}$$

(a) Is either estimator unbiased?

(b) Which estimator is better? In what sense is it better? Calculate the relative efficiency of the two estimators.

Exercise 7.23: Suppose that $\hat{\Theta}_1$ and $\hat{\Theta}_2$ are estimators of the parameter θ. We know that $E(\hat{\Theta}_1) = \theta$, $E(\hat{\Theta}_2) = \theta/2$, $V(\hat{\Theta}_1) = 10$, $V(\hat{\Theta}_2) = 5$. Which estimator is better? In what sense is it better?

Exercise 7.25: Let three random samples of sizes $n_1 = 8$, $n_2 = 14$, and $n_3 = 6$ be taken from a population with mean μ and variance σ^2. Let S_1^2, S_2^2, and S_3^2 be the sample variances. Show that $S^2 = (7S_1^2 + 13S_2^2 + 5S_3^2)/25$ is an unbiased estimator of σ^2.

Exercise 7.29: Data on the oxide thickness of semiconductor wafers are as follows: 425, 431, 416, 420, 421, 436, 418, 410, 431, 433, 423, 426, 411, 435, 436, 428, 411, 426, 407, 437, 422, 428, 413, 416.

(a) Calculate a point estimate of the mean oxide thickness for all wafers in the population.

(b) Calculate a point estimate of the standard deviation of oxide thickness for all wafers in the population.

(c) Calculate the standard error of the point estimate from part (a).

(d) Calculate a point estimate of the median oxide thickness for all wafers in the population.

(e) Calculate a point estimate of the proportion of wafers in the population that have oxide thickness of more than 430 angstroms.

Exercise 7.32: Two different plasma etchers in a semiconductor factory have the same mean etch rate μ. However, machine 1 is newer than machine 2 and consequently has smaller variability in etch rate. We know that the variance of etch rate for machine 1 is σ_1^2, and for machine 2, it is $\sigma_2^2 = a\sigma_1^2$. Suppose that we have n_1 independent observations on etch rate from machine 1 and n_2 independent observations on etch rate from machine 2.

(a) Show that $\hat{\mu} = \alpha\bar{X}_1 + (1-\alpha)\bar{X}_2$ is an unbiased estimator of μ for any value of α between zero and one.

(b) Find the standard error of the point estimate of μ in part (a).

(c) What value of α would minimize the standard error of the point estimate of μ?

(d) Suppose that $a = 4$ and $n_1 = 0.5n_2$. What value of α would you select to minimize the standard error of the point estimate of μ? How "bad" would it be to arbitrarily choose $\alpha = 0.5$ in this case?